Genetic Algorithm & Its Applications

Principles · Dynamics · Crossover · Use of Fuzzy Membership · Worked Example

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Soft Computing — GA Module

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Course Page:

https://www.gcjana.in/courses/shardauniversity/2502/PHD650

Outline

1. Basic Working Principle of a Genetic Algorithm
2. Dynamics of GAs in Optimisation
3. Crossover Techniques
4. Fuzzy Set: Triangular Membership Function
5. GA Worked Example: $f(x) = x^2$
6. Summary

What is a Genetic Algorithm?

Definition

A **Genetic Algorithm (GA)** is a search and optimisation heuristic inspired by Darwin's principle of *natural selection* and biological evolution.

Key biological analogy:

- **Chromosome** — encoded candidate solution
- **Gene** — one decision variable (bit)
- **Population** — set of current solutions
- **Fitness** — quality measure of a solution
- **Reproduction** — creation of offspring

Application domains:

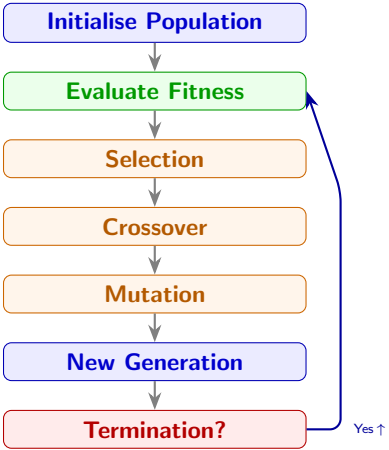
- Function optimisation
- Scheduling & routing (TSP)
- Machine learning & feature selection
- Game playing & robotics
- Engineering design

Key Idea

GAs search using *populations*, not single points — robust to noise and local optima.

GA Cycle Step by Step

1. **Initialise** random population of N chromosomes.
2. **Evaluate** each chromosome via fitness function f.
3. **Select** parents: fitter $\Rightarrow$ higher probability.
4. **Crossover**: exchange genetic material.
5. **Mutate**: randomly flip bits with prob. P_m.
6. **Replace** old population with offspring.
7. **Repeat** steps 2–6 until termination.



Exploration vs. Exploitation Trade-off

Exploration

- Searching **new** regions of the solution space
- High population diversity
- High mutation rate, low selection pressure
- Avoids premature convergence
- Most useful in **early** generations

△ Risk: may converge too slowly

Exploitation

- Refining **known** good solutions
- Low diversity; population converges
- High selection pressure, low mutation
- Fast convergence to best solutions
- Most useful in **later** generations

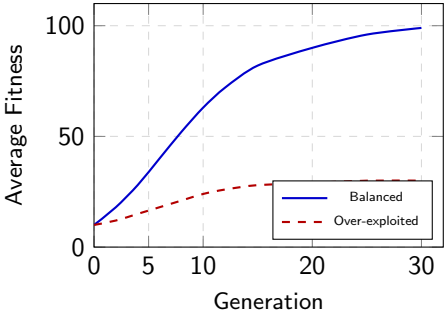
△ Risk: trapped in local optima

The Balance

Effective GAs balance both: explore broadly early, then exploit the best regions found.

How GAs Evolve Over Generations

- **Schema Theorem:** Short, low-order schemata with above-average fitness grow *exponentially* each generation.
- **Convergence:** Average fitness rises; variance falls as population clusters around fit solutions.
- **Premature Convergence:** Over-exploitation early ⇒ diversity loss and local optima.
- **Adaptive Operators:** Varying mutation/crossover rates maintain balance.



Generation Timeline

Gen 0: Random → 1–5: Rises → 6–20: Converges → 21+: Fine-tune

Single-Point & Two-Point Crossover

1. Single-Point Crossover

One cut point; left part from P1, right from P2.

P1	1	1	0	1	0	1
P2	0	1	1	0	1	0
O1	1	1	0	0	1	0
O2	0	1	1	1	0	1

cut

2. Two-Point Crossover

Two cut points; middle segment swapped.

P1	1	0	1	1	0	1	0
P2	0	1	0	0	1	0	1
O1	1	0	0	0	1	1	0
O2	0	1	1	1	0	0	1

P_1 P_2

Uniform Crossover & Other Techniques

3. Uniform Crossover

Each gene is independently selected from either parent via a random mask.

Row	Bit positions						
Parent 1	1	0	1	1	0	0	1
Parent 2	0	1	0	0	1	1	0
Mask	1	0	1	0	1	0	1
Offspring	1	1	1	0	0	1	1

Mask $= 1 \Rightarrow$ P1 — Mask $= 0 \Rightarrow$ P2

Arithmetic XO

Offspring $= \alpha P_1 + (1-\alpha) P_2$

For real-valued chromosomes.

Order XO (OX)

For permutation problems (TSP).

Preserves relative gene order.

PMX Crossover

Partially Mapped XO. Maps genes between two cut points.

Triangular Membership Function — Definition

Fuzzy Set "Temperature" — *Medium*

$a = 20°\text{C}$ (left), $b = 30°\text{C}$ (peak), $c = 40°\text{C}$ (right)

$$\mu(x) = \begin{cases} 0 & x \leq 20 \text{ or } x \geq 40 \\ \frac{x-20}{30-20} & 20 < x \leq 30 \\ \frac{40-x}{40-30} & 30 < x < 40 \end{cases}$$

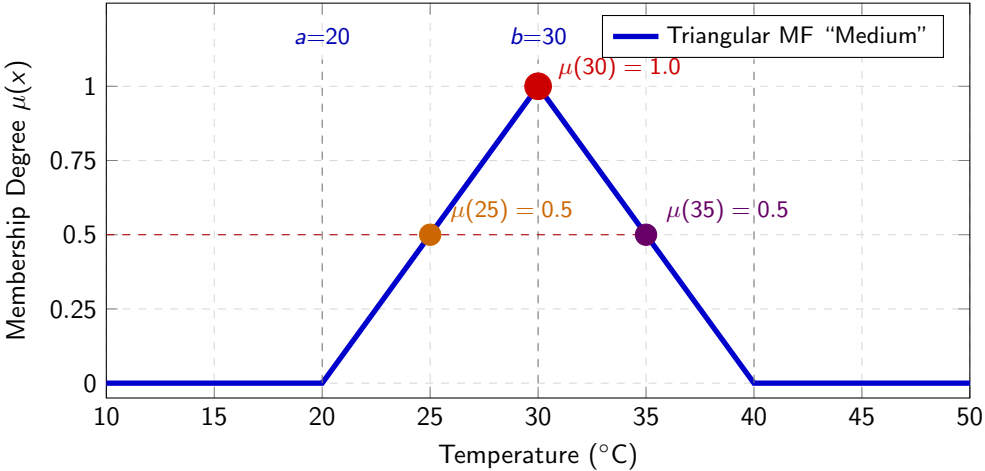
(a) Membership Calculations

x	Formula	$\mu(x)$
25°C	$\frac{25-20}{30-20} = \frac{5}{10}$	**0.5**
30°C	Peak ($x = b$)	**1.0**
35°C	$\frac{40-35}{40-30} = \frac{5}{10}$	**0.5**

(c) Interpretation

- $\mu(25) = 0.5$: 50% Medium — transitioning from cool to medium.
- $\mu(30) = 1.0$: Fully Medium — ideal peak temperature.
- $\mu(35) = 0.5$: 50% Medium — transitioning from medium to warm.

(b) Membership Function Graph

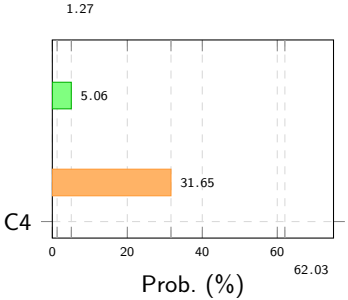


(a) Fitness Values & Selection Probability

Problem

Maximise $f(x) = x^2$, $x \in [0, 7]$, 3-bit binary encoding.

Chr.	Binary	x	$f(x)$	Sel. Prob.	Cumul.
C1	001	1	1	1.27%	1.27%
C2	010	2	4	5.06%	6.33%
C3	101	5	25	31.65%	37.98%
C4	111	7	49	62.03%	100%
Total			**79**	**100%**	



$$P(C_i) = \frac{f(x_i)}{\sum f(x)} = \frac{f(x_i)}{79}$$

(b) Roulette Wheel Selection

Cumulative Probability Ranges

Chr.	Range	$f(x)$
C1	0.00%–1.27%	1
C2	1.27%–6.33%	4
C3	6.33%–37.98%	25
C4	37.98%–100%	49

Spin Results

Spin	r	Selected
1	0.62	**C3** (6.33%–37.98%)
2	0.80	**C4** (37.98%–100%)
3	0.22	**C3** (6.33%–37.98%)
4	0.95	**C4** (37.98%–100%)

Mating Pairs

Pair 1: **C3** $\times$ **C4** (Spins 1 & 2)
Pair 2: **C3** $\times$ **C4** (Spins 3 & 4)

Note

C4 dominates selection with 62% probability, reflecting its superior fitness $f(7) = 49$.

(c) Single-Point Crossover (cut after 2nd bit)

Pair 1: C3 × C4

P1 (C3)	1	0	1
P2 (C4)	1	1	1
O1	1	0	1
O2	1	1	1

cut

Offspring Summary

	Binary	x	$f(x)$
O1	101	5	**25**
O2	111	7	**49**
O3	101	5	**25**
O4	111	7	**49**

Before Mutation

O1: 101 O2: 111 O3: 101 O4: 111

Pair 2 (C3×C4) gives identical offspring.

(d) Mutation & (e) New Population After One Generation

(d) Mutation $P_m = 0.1$

Last bit of O3 is flipped ($1 \rightarrow 0$):

Before [1] [0] [1] $x = 5$

↓ flip bit 3

After [1] [0] [0] $x = 4$

$f(4) = 4^2 = \mathbf{16}$ (was $f(5) = 25$)

(e) New Population

#	Source	Bin	x	$f(x)$	
1	P1 Pair 1	101	5	25	
2	P2 Pair 1	111	7	49	*
3	P1 Pair 2	101	5	25	
4*	P2 Pair 2 (mut.)	100	4	16	

Last bit flipped: $1 \rightarrow 0$

Generation Summary

Old avg = 19.75 New avg = **28.75**

Improvement: +45.6 % √

Summary

What We Covered

1. **GA Principle** — encode, evaluate, select, crossover, mutate; iterate.
2. **GA Dynamics** — fitness rises each generation; balance exploration & exploitation.
3. **Crossover** — single-point, two-point, uniform; different trade-offs.
4. **Fuzzy Membership** — triangular MF: $\mu(25^\circ)$=0.5, $\mu(30^\circ)$=1.0, $\mu(35^\circ)$=0.5; graded transitions.
5. **GA Worked Example** — $f(x)$=x^2: roulette favoured C3 & C4; crossover + mutation improved avg fitness by **+45.6%**.

Key Result

+45.6% fitness improvement in one generation — demonstrating the power of selection, crossover, and mutation.

Thank You

Any Questions?

"Evolution is the fundamental idea in all of life science and genetic algorithms bring that idea to computation."